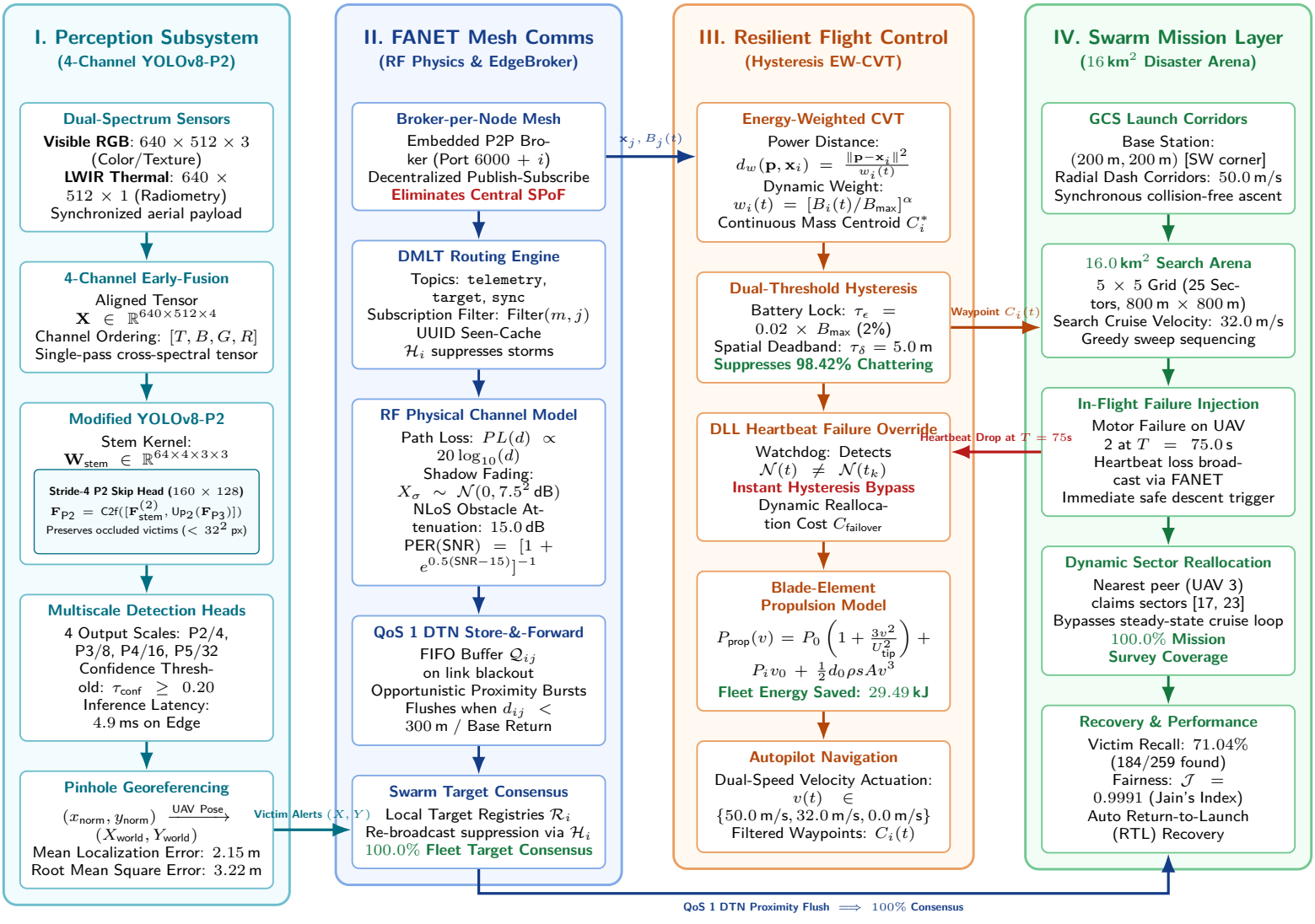

Figure 1: Integrated End-to-End System Architecture of FlyNetSim. The diagram details the four tightly coupled operational subsystems: (I) Multispectral Perception fusing RGB and LWIR thermal streams through a custom YOLOv8-P2 architecture with stride-4 skip connections; (II) Decentralized FANET Mesh featuring broker-per-node EdgeBroker, RF physics-informed channel modeling, and QoS 1 DTN store-and-forward queues; (III) Resilient Flight Control incorporating anti-chattering Dual-Threshold Hysteresis ($\tau_\epsilon, \tau_\delta$) for EW-CVT with automatic DLL failure bypass; and (IV) Swarm Mission Orchestration across a 16.0 km^2 post-disaster arena with Base Station corridors, mid-flight failover re-allocation, and optimal energy fairness ($\mathcal{J} = 0.9991$). Solid colored arrows denote forward data streaming and actuation commands; the dashed red arrow represents the asynchronous failure trigger.